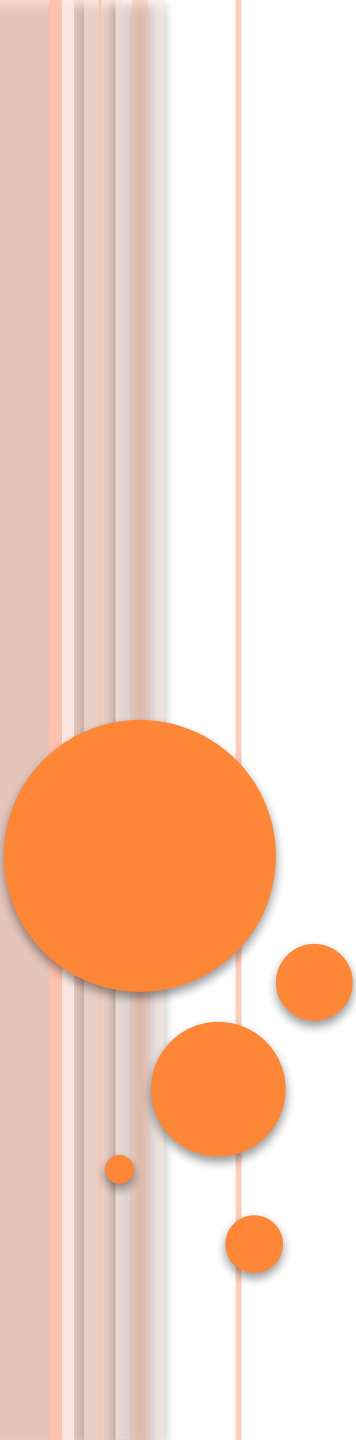




PHARMACOGNOSY

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2020**



UNIT I

Definition, History and Scope of Pharmacognosy

DEFINITION

- The word Pharmacognosy is derived from the Greek
 - "Pharmakon", meaning a drug or poison
 - "gignosis" meaning to acquire a knowledge of literally meaning "the entire knowledge of drug"
- Pharmacognosy is interdisciplinary, drawing on a broad spectrum of biological and socio-scientific subjects including

INTERDISCIPLINARY SUBJECTS...

- Subjects from which pharmacognosy provide and borrows knowledge from includes
 - Botany, ethnobotany, medical anthropology, marine biology, microbiology, herbal medicine, chemistry, biotechnology, Phytochemistry, pharmacology, pharmaceutics, clinical pharmacy and pharmacy practice

DEFINITIONS

- The American Society of Pharmacognosy defines pharmacognosy as:
 - ***“ the study of the physical, chemical, biochemical and biological properties of drugs, drug substances or potential drugs or drug substances of natural origin as well as the search for new drugs from natural sources ”***

RELATED TERMS

- Ethnobotany: It is a broad term referring to the study of plants that are used by humans
- Ethnomedicine: It refers to the use of plants by humans as medicine
- Traditional medicine: It is the sum total of all nonmainstream medical practices, usually excluding so called “western” medicine

RELATED...

- Phytotherapy: It is the study of the use of extracts from natural origin as medicines or health-promoting agents
- Phytochemistry: It is (from the Greek words phyton = plant and chemeia = chemistry) deals with the knowledge of natural products from plant origin exhibiting bioactivity

RELATED...

- Three primary roles for phytochemistry in the future can be presented:
 - Drug discovery.
 - Establishing the rationale for the successful employment of crude plant extracts
 - Development of plants as chemopreventive agents

RELATED...

- Natural products:
- they can be
 - I. Entire organism (plant, animal, organism)
 - II. Part of an organism (a leaf or flower of a plant, an isolated gland or other organ of an animal)
 - III. An extract or an exudate of an organism
 - IV. Isolated pure compounds

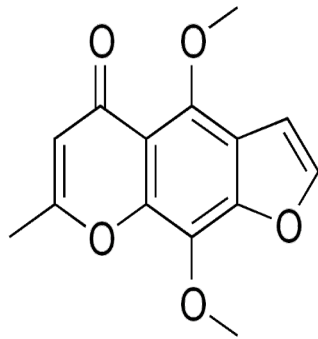
WHY DO WE STUDY PHARMACOGNOSY (NP)?

- Herbs and natural products offer a very good alternative source of drugs.
- (FDA, 1997) new approved drugs reported between 1983-1994 [Total 520]:
 - Drugs of natural origin: unmodified, semisynthetic from a natural source, or modeled on a natural product parent: 40%.

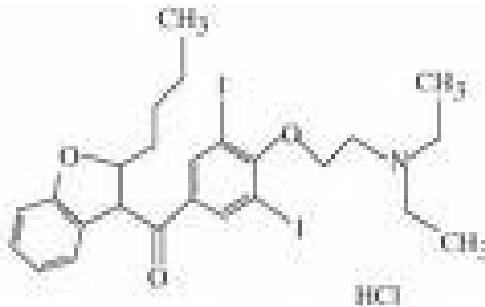
WHY...

- Naturally-derived drugs predominate in the area of:
 - Antibacterials (78%),
 - anticancer drugs (61%) and
 - other diseases (30-50%)

EXAMPLES OF NPs AND THEIR DERIVATIVES



Khellin



Amiodarone

WHY..

- Some natural products contain compounds that demonstrate little or no activity themselves but which can be modified
 - *Dioscorea* yam: molecule -> steroids
 - Soya: saponins -> steroids
 - *Baccatin III* → *Taxols*

- Taxol is a low yieldc cpd
- it is estimated that about 60% of all steroidal drugs are derived from
- progestrone
diosgenin

WHY...

- Their utility as prototypes or models for synthetic drugs possessing physiologic activities similar to the originals

WHY...

- Source of toxic molecules
 - To study the way the body responds to their pharmacological use
 - Investigating pharmacological mechanisms
 - Picrotoxin – nerve conduction
 - Carrageenan- anti-inflammatory

I. THE HISTORY OF NATURAL PRODUCTS IN MEDICINE

- A great proportion of the natural products used as drugs
- The study of drugs used by traditional healers is an important object of pharmacognostical research
- Sumerians and Akkadians (3rd millennium BC)

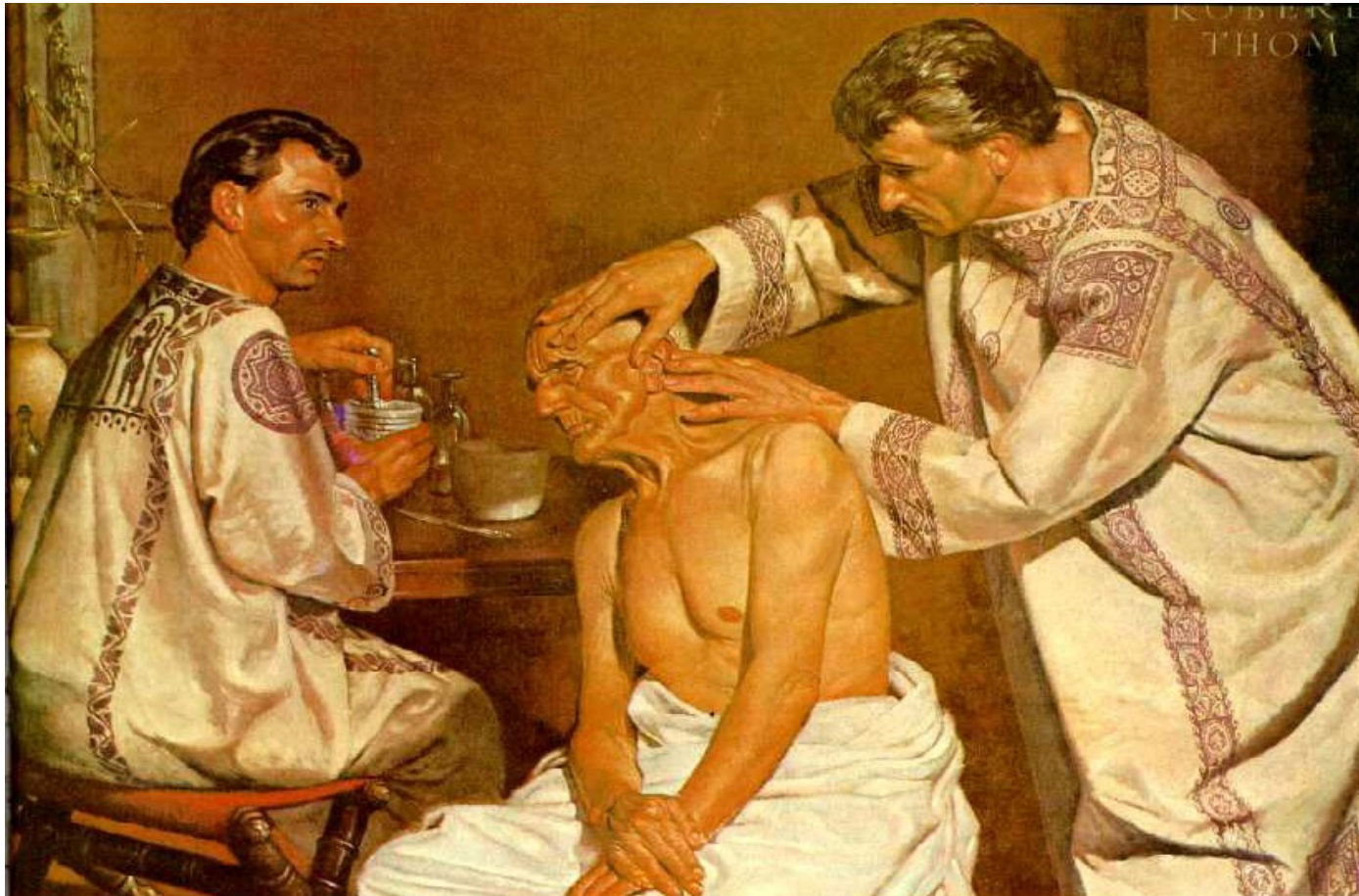


EGYPTIANS (*EBERS PAPYRUS*, 1550 BC)



AUTHORS OF ANTIQUITY

- **Hippocrates** (460-377 BC)-“The Father of Medicine”



DIOSCORIDES (40-80 AD)

“DE MATERIA MEDICA” (600 MEDICINAL PLANTS)



GALEN (131-200 AD),

- A Greek pharmacist-physician,
- described methods of preparing pharmaceutical formulations containing plant and animal drugs.
- These methods & his other observations on medicinal plants have been recorded in a books.
- The present day Galenical preparations or Galenicals are prepared according to those methods.

THE ISLAMIC ERA

IBN ALTABARI (770–850),

**FIRDOUS AL HIKMAH ("PARADISE OF WISDOM")-DISEASE,
TOXICOLOGY, ASTRONOMY, PHILOSOPHY)**



IBN SINA (980-1037)

"CANON OF MEDICINE".

Formulary, disease, philosophy, anatomy, physiology, cosmetics



IBN ALBITAR (1148-1197)

AL-JAME', A COLLECTION OF BASIC MEDICINES AND NUTRIENTS DERIVED FROM PLANTS, ANIMALS AND MINERALS



THE ERA OF EUROPEAN EXPLORATION OVERSEAS (16TH AND 17TH CENTURY)



THE 18TH CENTURY, PHARMACOGNOSY

- Johann Adam (1759-1809)
- Linnaeus (**naming and classifying plants**)
- At the end of the 18th century, crude drugs were still being used as powders, simple extracts, or tinctures

THE ERA OF PURE COMPOUNDS

A NEW ERA IN THE HISTORY OF MEDICINE)

- Isolation of morphine from opium
- Strychnine (1817)
- Quinine and caffeine (1820)
- Nicotine (1828)
- Atropine (1833)
- Cocaine (1855)

- In the 19th century, the chemical structures of many of the isolated compounds were determined
- In the 20th century, the discovery of important drugs from the animal kingdom, particularly hormones and vitamins.
- Microorganisms have become a very important source of drugs

SCOPE OF PHARMACOGNOSY

- Traditionally the scope of Pharmacognosy includes:
- Identifying biological sources of drugs.
 - Plants, Microorganisms (bacteria, fungus), Marine organisms & Animal products
- Assessing the efficacy and safety of herbal drugs or their constituents as therapeutic agents using scientific methods.

SCOPE..

- Modern concept in Pharmacognosy includes:
- securing regular supply of natural products:
 - cultivation
 - increasing yield
 - Semi-synthesis and synthesis? **Why???**
- ~ 100 species/day is lost i.e. loss of potential new medicines.

SCOPE...

- incorporation of a natural product in the system of modern medicine.

GENERAL PRINCIPLES OF BOTANY: MORPHOLOGY AND SYSTEMATICS

- *a botanical drug is a product that is either:*

Derived from a plant and transformed into a drug by drying certain plant parts, or sometimes the whole plant, or

- Obtained from a plant, but no longer retains the structure of the plant or its organs and contains a complex mixture of biogenic compounds (e.g. fatty and essential oils, gums, resins, balms)

BOTANICAL DRUGS..

◆ Plant organs

1. Aerial parts or herb (herba)
2. Leaf (folia)
3. Flower (flos)
4. Fruit (fructus)
5. Bark (cortex)
6. Root (radix)
7. Rhizome (rhizoma)
8. Bulb (bulbus)

BOTANICAL DRUGS...

- The large majority of botanical drugs in current use are derived from leaves or aerial parts.
- A plant-derived drug should be defined not only in terms of the species from which it is obtained but also the plant part that is used to produce the dried product. Thus, a drug is considered to be adulterated if the wrong plant parts are included (e.g. **aerial parts instead of leaves**)

MORPHOLOGY OF HIGHER PLANTS

I. Flower

- It is the essential reproductive organ of a plant.
- For an inexperienced observer, two characteristics of a flower are particularly noteworthy: the size and the color
- Although the flowers are of great botanical importance, they are only **a minor source of drugs** used in phytotherapy or pharmacy e.g. chamomile, *Matricaria recutita* L. (Asteraceae)

2. Fruit and seed

- The lower plants, such as algae, mosses and ferns, do not produce seeds
- ***Gymnosperm and Angiosperm***
- **Gymnosperm**: they are characterized by seeds that are not covered by a secondary outer protective layer, but only by the testa – the seed's outer layer
- **Angiosperm**: the seeds are covered with a specialized organ (the carpels) which in turn develop into the pericarp.

- Drugs from the fruit thus have to be derived from **an *angiosperm*** species
- Fruits and seeds have yielded important phytotherapeutic products, including:

- **Fruit**

Caraway, *Carum carvi* L. (Umbelliferae)

- **Seed**

(white) mustard, *Sinapis alba* L. (Brassicaceae)

3. Leaves

- The function of the leaves, as collectors of the sun's energy and its assimilation, results in their typical general anatomy with a petiole (stem) and a lamina (blade)
- A key characteristic of a species is the way in which the leaves are arranged on the stem, they may be:
 1. Alternate
 2. Distichous
 3. Opposite
 4. Decussate
 5. Whorled

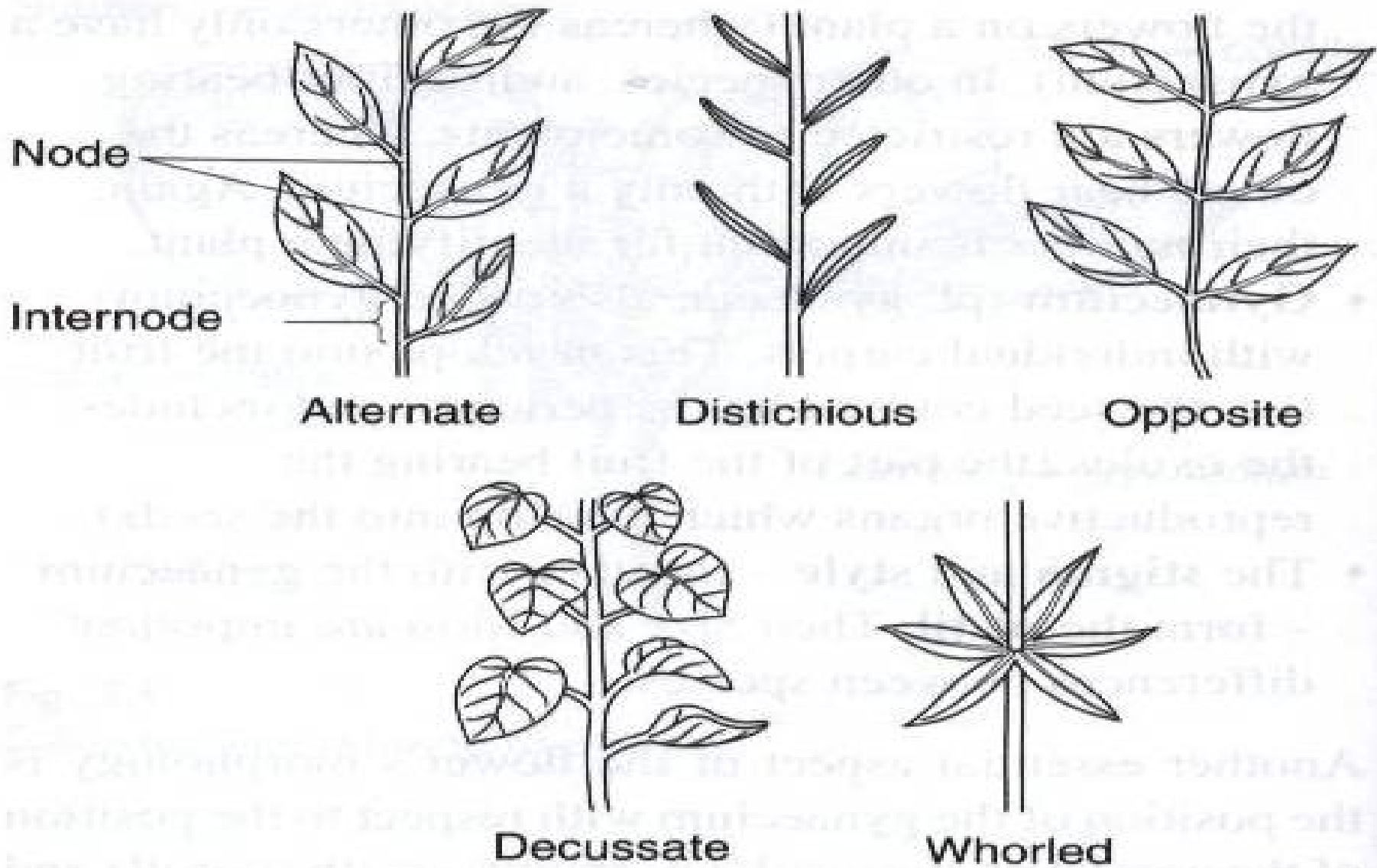


Fig. 3.3
Types of arrangements of leaves.

- The form and size of leaves are essential characteristics
e.g. oval, oblong, obovate, rounded, linear, lanceolate,
elliptic, spatulate, cordate, hastate or tendril
- The margin of the leaf is another characteristic feature
e.g. entire, serrate, dentate, sinuate, ciliate or spinose

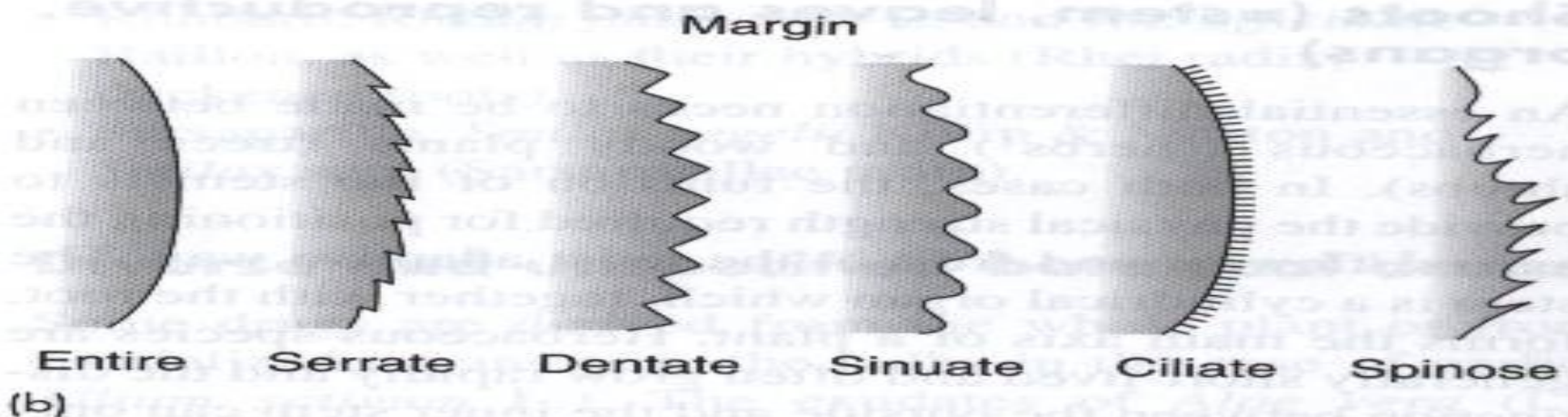
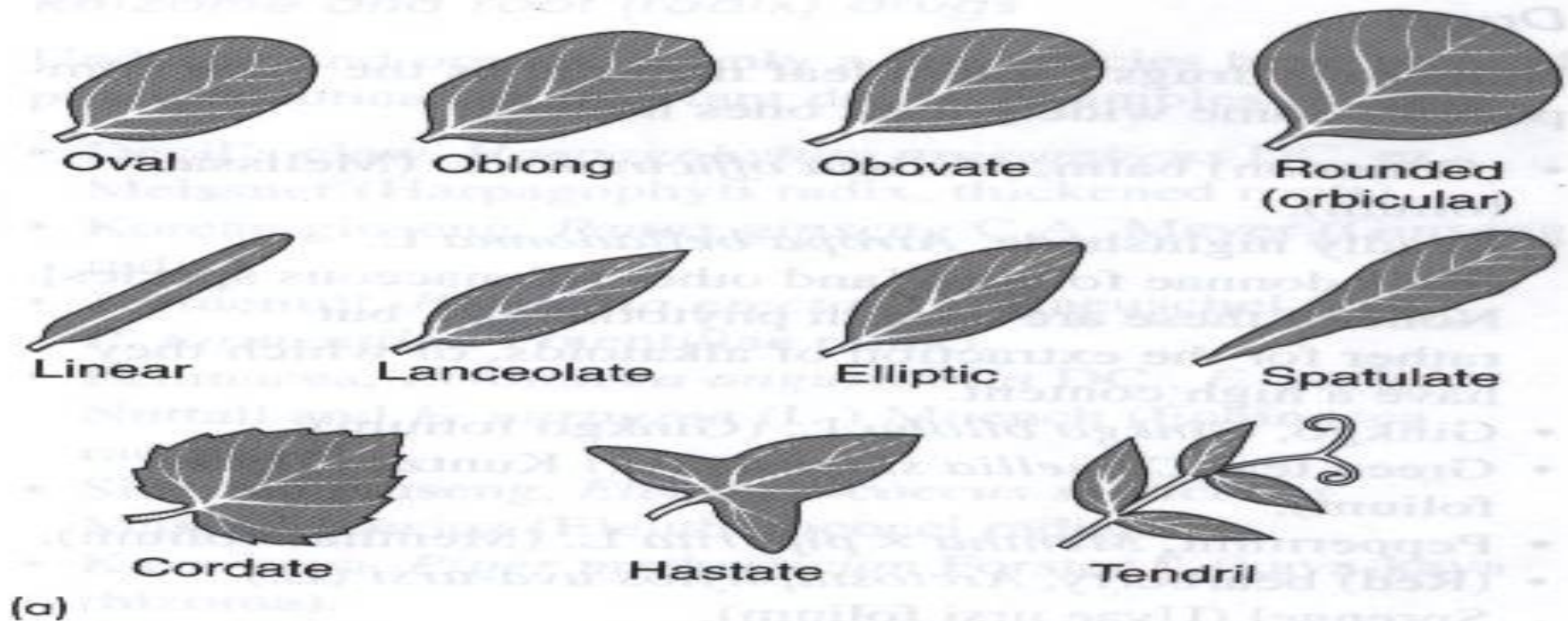


Fig. 3.4

(a) Characteristic shapes of leaves. (b) Characteristic margins of leaves.

- Numerous drugs contain leaf material as the main component.

e.g.

- Deadly nightshade, *Atropa belladonna* L. (Solanaceae)

4. Bark

- The bark as an outer protective layer frequently accumulates biologically active substances e.g.

Red cinchona, *Cinchona succirubra* L.

(Rubiaceae)

5. Rhizome and root drugs

- Underground organs of only a few species have yielded pharmaceutically important drugs e.g.
 1. Sarsaparilla, *Smilax regelii*
(Smilacaceae)
 2. Korean ginseng, *Panax ginseng*
(Araliaceae)

6. The bulbs and exudates

1. Garlic, *Allium sativum* L. (Liliaceae)
2. *Aloe vera* L. (Asphodelaceae)

Thank you